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# Early ageing detection of battery cells in battery management system

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In this Letter, a battery state of health monitoring technique suitable for battery management systems is presented. This method analyses switching and battery transitions in the battery management system topologies to distinguish any early degraded battery cell in the battery strings. The proposed technique processes the voltage of the system, captured by an already employed sensor during the transition of the states. To verify the performance of the proposed technique, some experiments have been carried out on a typical charge balancing circuit.

*Introduction:* One of the main challenges of humanity in the current days is energy resources. Fossil fuels, as a dominant energy source, produce an excessive amount of greenhouse gases. To limit the dependence of energy demand from fossil fuels, usage of renewable energies such as solar energy, wind energy, and hydro energy have become so trend currently. As these energy resources have an intermittent nature, energy storage systems (ESSs) should be employed alongside these sources to enhance their utilisation factor. Different ESSs such as flywheels, pumped storage systems, compressed air ESSs, and batteries have been utilised in industries. Among them, electrochemical batteries have been more attractive, despite some inherent limitations.

Different types of batteries are currently available, which lithium-ion (Li-ion), nickel–cadmium (Ni–Cd), lead–acid, and nickel–metal hydride are among the most utilised battery types. Because of high energy density, low self-discharge, low maintenance, and high current supply capability, Li-ion batteries are attractive choices in electrical vehicles (EVs), aerospace, and mobile applications. However, this battery type has some drawbacks like high-cost, the need of protection circuitry, and early ageing effects. Since voltage and stored energy of a single Li-ion battery cell is very low, many cells should be connected in series and parallel to create a battery string with the required voltage and energy level for an EV or a satellite. Produced battery cells, even in the same production procedure, may have some difference in their specifications. Such differences could negatively affect charge equalisation procedure and may result in early degradation of the cells in the long-term. To have charge balancing even with some minor differences, battery management systems (BMSs) are usually utilised, especially in applications with a high degree of importance.

BMSs have different functions like charge balancing, state of health (SOH) estimation, battery condition monitoring, and charging/discharging rate supervision. Many different charge balancing methods and reconfigurable battery packs have been presented in the literature [1–6]. In [1], a charge balancing method using an additional battery has been presented. In this technique, the unequal battery is disconnected from the battery string to start the charge balancing procedure. In [2], a battery switching layout has been suggested for reconfiguration of the batteries, utilised in different applications such as charge balancing. In this structure, each battery can be connected in series, parallel, or even bypassed. In [3], a reconfigurable battery pack has been proposed. In this structure, every battery cell and battery module have a relevant switch, utilised for reconfiguration of the battery pack as the load demand changes. In [4], different BMSs with reconfigurable battery packs have been reviewed. Various targets such as charge balancing, dynamic load supply, and being fault-tolerant can be carried out by these structures. A BMS based on utilising two different types of battery packs has been presented in [7]. In this BMS, considering different operation modes of the EV, the suitable battery type is connected to the system.

Another main objective of the BMSs is battery condition monitoring and SOH estimation [8–11]. In [9], two voltage pulses are applied to the system and based on the response, SOH of the batteries are estimated. In [10], SOH of the batteries is determined by charging and discharging battery cells for multiple times at a constant rate and analysing the charging/discharging trend. An SOH estimation based on a battery equivalent model using a Kalman filter has been presented in [11]. In this method, SOH of the batteries is estimated by identifying the internal resistance of the batteries in charging and discharging procedures. Although charge balancing of the battery cells is followed up by the BMSs, this function could not be realised efficiently when some early degraded cells exist in the battery string. Early ageing of any cell should be distinguished as soon as possible and it should be removed from the battery string to guarantee the performance of the BMS. Since the early degraded battery cells find different internal specifications, their step response differs in comparison with the healthy condition. In this work, this change is utilised as a signature for early ageing detection of the cells.

Many of the BMSs utilise switching components in their structure to reconfigure battery cells. In all these methods, connection and disconnection of the switches impose a state transition on the battery system. As the degraded batteries have different characteristics, their step response is somewhat different from the response of the other healthy battery cells. These differences are utilised for distinguishing a degraded battery in a battery string. Our experiments showed that there are some distinct dynamics in the voltage signal of the BMS when an early degraded battery is connected or disconcerted to the system.

In this Letter, a battery cell monitoring approach based on frequency analysis of the BMS voltage is presented. The voltage of a typical BMS is monitored when all the cells are in healthy condition and when some of them are in early ageing condition. The technique utilises just the already installed voltage sensors in the system. Frequency components of the dynamics relevant to the early degraded cells are derived using FFT. Based on the components, a criterion is proposed for the detection of early degraded cells. Square of difference between the consecutive FFT samples is calculated as the criterion. Some experiments are carried out on a prototype charge equalisation circuit with and without early degraded battery cell. The results confirm the distinctive signature of both conditions, which can be discriminated by the proposed criterion.

In the rest of this Letter, the working principle of the proposed method is presented in section *Proposed Battery Cell Monitor Technique*. Some experimental results are presented in *Experimental Results* section to verify the performance of the proposed technique. Finally, the main conclusions are presented in the *Conclusion* section.

*Proposed battery cell monitor technique:* Li-ion batteries have different electrical models. One of the utilised battery models is first-order RC model [12]. This model includes a voltage source, a discharge resistor for modelling the internal resistor, and a parallel RC branch to model voltage dynamics of the battery, as shown in Fig. 1. In this figure, $R_d$ models the internal resistor of the battery cell, $C_P$ and $R_P$ model the dynamic response of the system, $v_{OC}$ is the open-circuit voltage of the battery, $v_p$ is the dynamic voltage of the battery cell, and $v_t$ is terminal voltage of the battery cell.

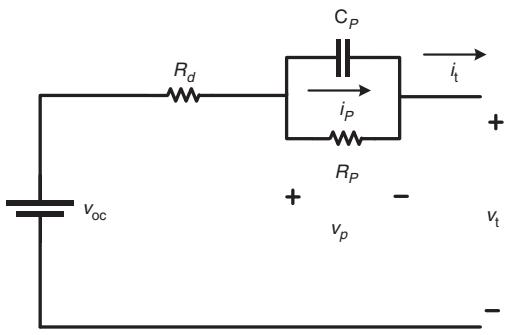



**Fig. 1** *Single RC model of a Li-ion battery*

The governed equations on the model are as follows:

$$v_t = v_{OC} - i_t \times R_d - v_p \tag{1}$$

$$v_p' = \frac{i_t}{C_p} - \frac{v_p}{C_p \cdot R_p} \tag{2}$$

using (1) and (2), the terminal voltage of the battery cell is calculated as follows:

$$v_t(t) = v_{OC} - R_d \cdot I - R_P \cdot I(1 - e^{-(t/\tau_p)}) \tag{3}$$

where $\tau_p = (1/(R_P \cdot C_P))$

As shown in (3), the terminal voltage of the battery cell includes a DC term and a decaying exponential term. As the internal characteristics of

the battery cell change, the dynamic response of the terminal voltage also changes. By analysing the frequency content of the system after any step response, discrepancies between the battery cells can be distinguished.

In the proposed technique, the voltage of the battery cells is monitored, continually. Whenever a transition state like connection or disconnection of a battery cell is applied to the battery string, impulse response of the system is analysed for extracting the signatures. As the dynamics of the transition states are very quick, the required time interval for recording the voltage signal is very short. After the recording period, the frequency content of the signal is extracted using FFT. It will be confirmed that the frequency contents of the early degraded and sound battery cells have some discrepancies. By analysing the frequency components, early ageing of the battery cells can be determined. FFT of the signal is calculated as follows:

$$Y_K = \sum_{n=0}^{N-1} x_n \cdot \left[\cos\left(\frac{2\pi kn}{N}\right) - \mathrm{j} \cdot \sin\left(\frac{2\pi kn}{N}\right)\right] \quad (4)$$

where $x_n$ is the input signal, $N$ is the period of the input signal, and $K$ is an integer number which denotes the harmonic order. Amplitude spectrum is the absolute value of the $Y$.

*Experimental results:* In this section, some experimental results are presented, which have been carried out on a typical charge equalisation circuit. The charge equalisation technique is based on connection and disconnection of battery cells. In the experiments, healthy and early degraded conditions of the battery cells are investigated. The laboratory test bench of the system is shown in Fig. 2. In the test bench, TMS320F28335 was employed for control and recording the voltage. This controller measures the voltage of the battery cells and captures the voltage of the battery string during any transition state. The voltage waveform is continually captured for 1-min time intervals. The captured data is overwritten to reduce the required memory. Upon a battery transition is carried out by the BMS, the recorded data will be stored for further analysis. The utilised battery string consists of three healthy battery cells and one early degraded battery cell.

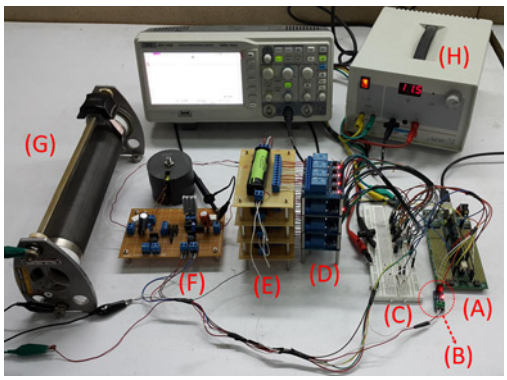



**Fig. 2** *Laboratory test bench (A) TMS320F28335 controller, (B) ACS712 current sensor, (C) Interface breadboard, (D) Charge balancing modules, (E) Battery pack, (F) Buck/boost converter, (G) Load, (H) DC source*

Two scenarios are considered. In the first scenario, one of the healthy cells is contributed in the transition state. In Fig. 3, the voltage of the system in this condition is shown. In the transition procedure, the voltage of the battery reaches zero for a short time interval (0.005 s) at the moment of the transition. For a very short time interval, the voltage of the battery string becomes negative which is because of the ringing effect. This phenomenon is mainly caused by the stray inductances and capacitances of the elements. Such sharp variation can be filtered in the analysis. In Fig. 4, frequency components of the signal, derived by FFT are presented. As shown in this figure, the frequency components between 0–150 Hz have higher amplitude compared with the other components. Square of the difference between the consecutive FFT samples is utilised as the criterion for detection of the degraded battery cells, named early degradation index (EDI), hereafter. EDI of a signal is calculated as follows:

$$S_i = (A_{i+1} - A_i)^2 \quad (5)$$

where $A$ is the amplitude spectrum of the signal.

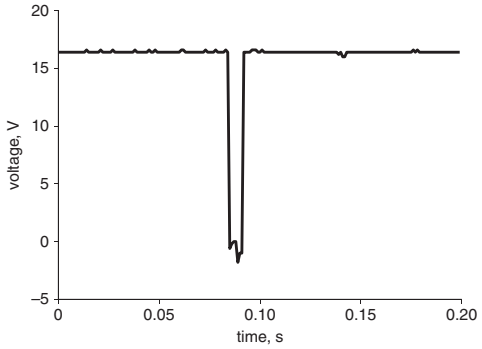



**Fig. 3** *Voltage response of the system during the transition in the first scenario*

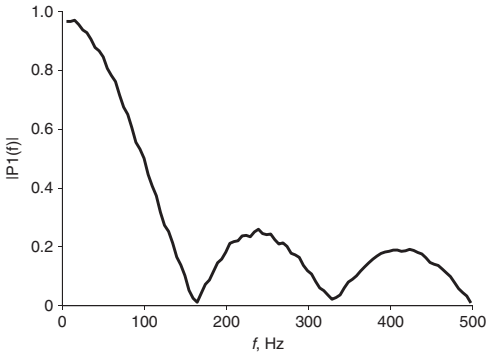



**Fig. 4** *Amplitude spectrum in the first scenario*

This calculated EDI for the first scenario is shown in Fig. 5. It is observed that the value of EDI is lower than 0.005 for all the components.

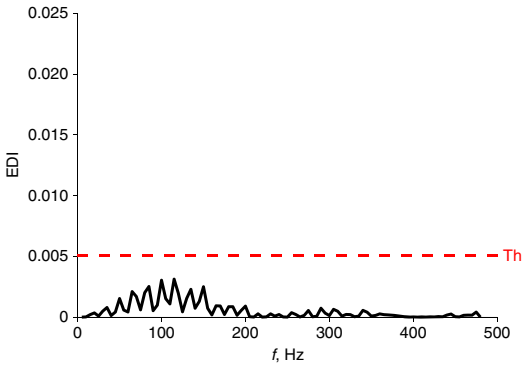



**Fig. 5** *EDI variation in the first scenario*

In the second scenario, an early aged battery cell is contributed in the transition state. Variation of the voltage in this condition is shown in Fig. 6. It is observed that the voltage response of the degraded battery includes low-frequency content after the transition in comparison with the healthy battery cell. Here, only one typical result is presented for each scenario. However, many experiments were carried out for both the scenarios, in which a similar variation is observed in the corresponding voltage signal. Amplitude spectrum of the voltage harmonic components in this scenario is shown in Fig. 7. As observed, some deviations are visible in the low-frequency band of the spectrum. Variation of EDI for this condition is depicted in Fig. 8. In addition to the significant overshoot observed at frequency band lower 50 Hz, some deviations are also observed at higher frequency bands. These observable discrepancies can be considered for detection of any early degraded battery cell. Here, EDI is compared with a threshold equal to 0.005 to detect any early degraded battery cell. In Figs. 5 and 8, this threshold is depicted by Th in red colour. The value of the threshold was derived by trial and error approach for the under-studied system. However, this threshold may be different in the other structures.



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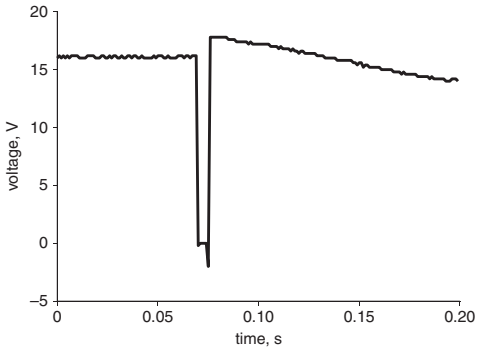



**Fig. 6** *Voltage response of the system during the transition in the second scenario*

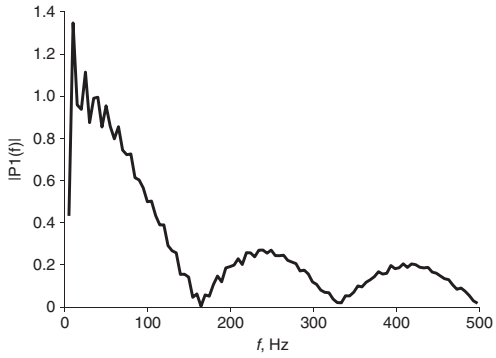



**Fig. 7** *Amplitude spectrum in the second scenario*

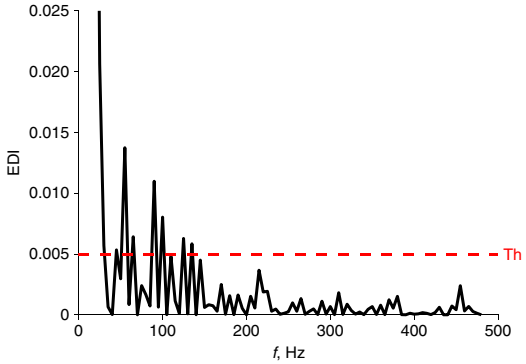



**Fig. 8** *EDI variation in the second scenario*

*Conclusion:* In this Letter, a battery cell monitoring technique was presented, which can be integrated with BMSs. This technique requires no additional sensors. This method analyses the transition dynamics of the normal operation of the charge balancing circuit. As the early degraded battery cells have different internal characteristics, their step response to the transitions is different from the healthy battery cells. It was concluded that there are some low-frequency dynamics in the voltage of the system when an early degraded battery cell is contributed in the transition. Using FFT and difference between their consecutive samples, a criterion was defined for detection of the early degraded battery cell. Some experiments were carried out for transition states of healthy and early degraded battery cells. By analysing the captured results, it was confirmed that the proposed method could discriminate both the condition, straightforwardly.

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One or more of the Figures in this Letter are available in colour online.

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